

CNC-X100 Vertical Machining Center Operation Manual

Product: CNC Machine | Model: CNC-X100 | Document Type: OPERATION_MANUAL

1. System Overview & Control Panel Usage

The CNC-X100 is a high-precision 3-axis vertical machining center engineered for heavy-duty production milling and drilling operations in industrial enterprise facilities. The machine is controlled via a 15-inch Touchscreen HMI panel running the ProductAssist OS interface.

The control panel consists of the primary touch display, emergency stop button, cycle start/hold buttons, manual pulse generator (MPG), mode selector dial (AUTO / MDA / JOG), and spindle speed override dial (0% to 150%).

- Touchscreen HMI: Displays real-time axis positions (G54-G59), active modal G-codes, tool offset table, coolant flow rate telemetry, and system alarm logs.
- Cycle Start Button (Green): Initiates automatic execution of loaded G-code programs.
- Feed Hold Button (Red): Pauses axis feed motion immediately without stopping spindle rotation or turning off coolant flow.
- Spindle Speed Override Dial: Allows real-time operator adjustment of spindle RPM between 0% and 150% during cutting cycles.

2. Network Configuration & Connectivity

The CNC-X100 features dual 1000BASE-T Ethernet ports located on the rear electrical cabinet door for integration with enterprise shopfloor networks and ProductAssist AI monitoring agents.

To configure network settings, navigate to Menu > Settings > System > Network Configuration on the control panel:

- IP Allocation: Static IP (Recommended for automated telemetry monitoring).
- Default Static IP Address: 192.168.1.150 | Subnet Mask: 255.255.255.0
- Gateway Address: 192.168.1.1 | DNS Server: 192.168.1.10
- Modbus TCP (Port 502): Transmits real-time sensor telemetry including spindle load, coolant pressure, and axis vibration to PLC monitoring hubs.
- OPC UA Server (Port 4840): Provides standardized machine state, operational mode, and fault alarm history to shopfloor SCADA applications.
- MTConnect Agent (Port 5000): Streams XML execution metrics and part completion counts for manufacturing execution tracking.

3. Pre-Start Inspection & Daily Startup Procedure

Operators must complete the standard 5-step daily startup routine prior to machining production parts:

- Step 1: Perform visual pre-start check of machine enclosure, clean chips from way covers, and verify coolant fluid level sight glass reads above 75% capacity.
- Step 2: Turn ON main power circuit disconnect switch located on the rear electrical cabinet.
- Step 3: Release the red mushroom Emergency Stop button on the operator panel by twisting clockwise.
- Step 4: Press the Power On button on the control panel and wait 45 seconds for system boot.
- Step 5: Perform Axis Homing routine by selecting HOME mode and pressing CYCLE START to establish reference zero for X, Y, and Z axes (G28).

4. Machine Operation, Program Execution & Tooling

Standard workpiece setup, tool management, and program execution procedure:

- 1. Load G-code program via USB drive or network share into memory location /NC_PROG/PART_01.NC.
- 2. Mount workpiece securely in pneumatic vise and probe work coordinate origin using edge finder or touch probe (G54).
- 3. Perform tool length offset measurement using laser tool setter for active tools in 24-pocket BT40 ATC magazine.
- 4. Select AUTO mode, enable single-block mode for initial dry-run verification, and press CYCLE START.
- 5. Monitor spindle load meter and coolant flow rate during initial cutting pass.

5. Shutdown Procedures & Alarm Recording

Standard Shutdown Procedure:

- Move axes to safe home position (G28 X0 Y0 Z0). Stop spindle and turn OFF coolant pump.
- Navigate to Menu > System > Shutdown on HMI and select Confirm. Turn OFF main electrical disconnect breaker.

Alarm Recording Protocol:

- Whenever an alarm code (e.g. E105, E210, E315, E420) triggers during operation, the operator must log the alarm code, active tool number, G-code line, and axis positions into the shift maintenance log.

WARNING / SAFETY PRECAUTION:

In case of tool crash, severe vibration, or coolant line rupture, hit the red Emergency Stop button immediately. Never reach inside enclosure while spindle is rotating.